

Tai Wong

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Education

University of Illinois at Urbana-Champaign

Champaign, IL

Bachelor of Science in Computer Science and Bioengineering / GPA: 4.0 / Dean's List Expected Graduation: May 2028

Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Discrete Structures, Systems Programming

Experience

Founder & Lead Developer

2025 – Present

SkyHub — Full-Stack iOS App, 15,000+ Monthly Active Users

SwiftUI, FastAPI, PostgreSQL, Docker

- Architected and shipped a full-stack iOS app serving 15,000+ MAU, building a 42-endpoint FastAPI backend with PostgreSQL, background workers for live data aggregation, and automated health checks behind a Cloudflare Tunnel for zero-downtime public access.
- Delivered sub-second API response times by designing async task queues for heavy computations and containerizing all services with Docker for seamless scaling and continuous deployment.

Research Assistant

2025 – Present

Dream Lab, UIUC — Multi-Agent AI Systems & LLM Trustworthiness

Champaign, IL

- Designed and deployed a multi-agent LLM system coordinating specialized agents across three heterogeneous data sources (PubMed, NIH RePORTER, ClinicalTrials.gov) to automate biomedical research-gap identification, achieving best-in-class precision/recall on internal benchmarks.
- Built and scaled a Neo4j knowledge graph to 8M+ nodes by engineering ETL pipelines that ingest, deduplicate, and cross-link records from federal research APIs, enabling sub-second graph traversal for downstream LLM-driven analysis.
- Reduced LLM-fabricated outputs by 87% by engineering a deterministic code-first verification layer with grounding checks, improving reliability of multi-agent recommendations in production.

Research Collaborator

2022 – Present

Mattis Lab, UCSF — Trustworthy LLM Extraction & Computational Biology

San Francisco, CA

- Built and published ProtoHep (doi:10.5281/zenodo.19775274), the largest structured hepatocyte differentiation protocol database (878 records), via a multi-agent LLM pipeline automating extraction from PDF ingestion through schema validation; accepted for poster at the 2026 SLAM Conference.
- Achieved 0% hallucination rate at >0.8 extraction confidence by implementing a hybrid verification system combining text-matching heuristics with sandboxed code execution to validate every extracted field against source documents.
- Engineered an automated cross-document linking system using deep-agent web search to resolve references, accession numbers, and supplementary data across papers, eliminating a previously manual reconciliation step.

Projects

BioQuarry — Multi-Agent Biomedical Data Extraction Platform

2025 – Present

Python, LangGraph, Pydantic, asyncio, Neo4j, marker-pdf, NCBI E-utilities, React, TypeScript

- Architected a domain-agnostic async pipeline orchestrating search, triage, and multi-pass extraction agents via LangGraph with configurable schemas and human-in-the-loop approval gates, scaling to 100 papers in under 60 seconds via asyncio parallelism.
- Enforced end-to-end data integrity through a strict type boundary (Draft → GroundedValue) routing every extracted field through one of six deterministic verification paths before commit, with failures surfaced as structured error objects.

MeshNL — Natural Language to MeSH Semantic Search

2025

Python, PyTorch, BiomedBERT, BioLORD, HuggingFace Transformers, contrastive learning

- Fine-tuned a dual-encoder retrieval model (BiomedBERT + BioLORD) with contrastive loss for semantic search over 30,000+ MeSH ontology terms, achieving 95.8% macro recall across 15 category branches.
- Trained a BioLORD encoder with InfoNCE loss and hard-negative mining on 1M query-term pairs, delivering a 2.2× Recall@50 improvement over the zero-shot baseline.

Folio — RAG Toolkit for iOS & macOS (Open-Source Swift Package)

2025

Swift, SQLite FTS5, BM25, vector embeddings, Apple Foundation Models, SPM

- Engineered and published a zero-config RAG library as an installable Swift Package combining BM25 (SQLite FTS5) with vector embeddings for hybrid retrieval, with on-device LLM support via Apple Foundation Models (iOS 18+) and adapters for Ollama and any OpenAI-compatible API.

Technical Skills

Languages: Python, Swift, TypeScript, Java, C++, SQL, JavaScript

AI & ML: PyTorch, HuggingFace Transformers, LangGraph, RAG, Embeddings, Fine-Tuning, Contrastive Learning, LLM Evaluation, Prompt Engineering, Agentic Workflows, Multi-Agent Systems, CNNs, NLP, Knowledge Graphs, Neo4j

Data & Infrastructure: PostgreSQL, SQLite, MongoDB, Docker, FastAPI, AWS, Pydantic, asyncio, REST APIs, GitHub Actions (CI/CD), Linux

Dev Practices: Git, Unit & Integration Testing, Code Review, Agile/Scrum, System Design